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Introduction to Message Passing Interface (MPI)

Lesson 6

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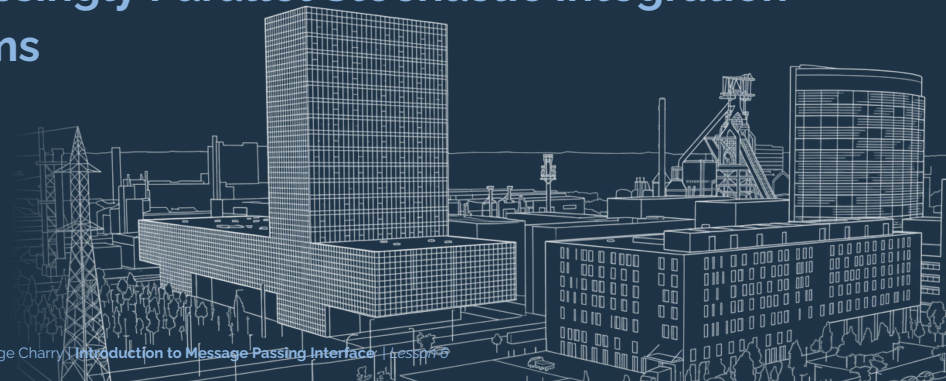
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21st of October 2026



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Embarrassingly Parallel Stochastic Integration algorithms



Embarrassingly Parallel

Embarrassingly Parallel Stochastic Integration algorithms

An **embarrassingly parallel workload** is a problem that can be divided into multiple completely independent tasks that require little to no inter-process communication or synchronization during execution.

In particular they are found in **Monte Carlo simulations** and **Stochastic Differential Equations (SDEs)**.

Both types of applications are in fact based on parallel **random walks** through which the properties are **sampled**:

- Each MPI process initializes and calculates the dynamics of its local trajectory (or subset of trajectories) executed entirely using local memory.
- The parallel sampling is distributed within the MPI processes to guarantee precise load balance.
- Communication occurs only once at the very end using a collective reduction (MPI_Reduce or MPI_Allreduce) to sum or average the path estimates.
- This guarantees both nearly perfect Speed Up and efficiency.

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Stochastic Integration



Numerical evaluation of π

Stochastic Integration

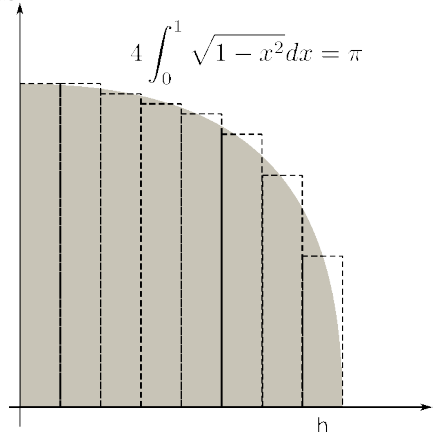
The 'easy' way of computing the value of π from a finite integral can be done by considering for example the area of the semi-circle with radius r that can be written as:

$$\int_{-r}^{r} \sqrt{r^2 - x^2}dx = \frac{1}{2}\pi r^2 \qquad (1)$$

and assuming for simplicity $r = 1$ we have that

$$2\int_{-1}^{1} \sqrt{1 - x^2}dx = \pi. \qquad (2)$$

thus π can be obtained through the trapezoidal rule with an error of order $\mathcal{O}(h^2)$ where h is the length of the segments in which the interval of integration is divided.



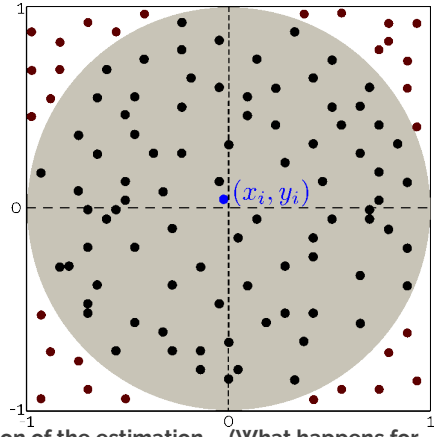
Stochastic evaluation of π

Stochastic Integration

The classical illustrative example for the stochastic integration is the evaluation of π through the extraction of random numbers.

The idea behind the stochastic evaluation of π is simple. Consider a circle of unitary radius, then:

1. Extract $\mathcal{M}$ random points defined by a couple of uniform random numbers $x, y \in [-1 : 1]$ count the number m random points for which $x^2 + y^2 \leq 1$
2. Given that $\frac{A_{\text{circle}}}{A_{\text{box}}} = \frac{\mathcal{N}_{\text{circle}}}{\mathcal{N}_{\text{square}}}$.
   π will be equal to the ratio between the total number of points extracted and those inside the circle times the area of the box:
   $\pi = 4\frac{m}{\mathcal{M}}$.



Convergence is much slower than the numerical version of the estimation ... (What happens for multi-dimensional integrals?)

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Monte Carlo integration



Monte Carlo Integration

Monte Carlo integration

Let us consider the integral, in the interval $[a, b]$ of a general function $f(x)$.

$$I = \int_a^b f(x)dx \tag{3}$$

By extracting, with uniform probability $\mathcal{N}$ variables x_i in the interval $[a, b]$ we can estimate an integral stochastically as

$$I \approx \frac{(b-a)}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} w_i f(x_i) \tag{4}$$

where $w_i = 1$. This estimation will have an error that decreases has the square root of the number of samplings $\mathcal{N}$

$$\sigma[I] = \sqrt{\frac{Var[f(x)]}{\mathcal{N}}} \tag{5}$$

and is proportional to the variance of $f(x)$ in the intervale $x \in [a, b]$ defined as

$$Var[f(x)] = \frac{1}{(\mathcal{N}-1)} \sum_{j=1}^{\mathcal{N}} \left(\frac{(b-a)}{\mathcal{N}} w_j f(x_j) - \frac{(b-a)}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} w_i f(x_i) \right)^2 . \tag{6}$$

Error convergence between stochastic and numerical integration

Monte Carlo integration

The fact that the error is proportional to the inverse of the square root of the sampling, i.e. $\sigma[I] \propto \frac{1}{\sqrt{N}}$ is to be expected from the central limit theorem.

This scaling is clearly unfavorable compared with standard quadrature methods, such as the Trapezoidal Rule, that use equidistant values for x_i and have an error of the order $\mathcal{O}(h^k) = \mathcal{O}(\frac{1}{N^k})$ where k is associated with the order of the polynomial approximating the function ($k = 2$ for the trapezoidal Rule).

Monte Carlo integration is more efficient in multi-dimensional integrations.

Assuming we have to integrate in an hypercube of size L^d volume where d is the dimensionality of the system, and we divide each dimension in intervals of h length, such that the number of points will be $N = (L/h)^d$ the error will scale as $\mathcal{O}(N^{-k/d})$.

The error in the Monte Carlo integration will still be of order $\mathcal{O}(\frac{1}{\sqrt{N}})$.

Thus Monte Carlo is more efficient if $d > 2k$. If we thus have a system of N quantum particles $d = 3N$.

Importance Sampling

Monte Carlo integration

In order to reduce the $Var[f(x)]$ of the uniform sampling, we can chose points that are concentrated in the region where $f(x)$ is large in absolute values.

Let $\Pi(x)$ be a positive function in the interval of integration $[a, b]$, chosen so that $f(x)/\Pi(x)$ is nearly constant in the interval and so that $\int_a^b \Pi(x)dx = 1$.

We then write the integral over $f(x)$ as:

$$I = \int_a^b f(x)dx = \int_a^b \frac{f(x)}{\Pi(x)}\Pi(x)dx. \tag{7}$$

The stochastic integration now requires the extraction of x_i points according to the distribution $\Pi(x)$ so that

$$I \approx \frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} \frac{f(x)}{\Pi(x)} \tag{8}$$

with an error $\sigma_I = \sqrt{\frac{Var[\frac{f(x)}{\Pi(x)}]}{\mathcal{N}}}$ that decreases as the square root of the variance of the ratio.

Variance convergence in importance sampling

Monte Carlo integration

While in the simple sampling the variance is in fact given by

$$Var[f(x)] = \frac{1}{(\mathcal{N}-1)} \sum_{j=1}^{\mathcal{N}} \left(\frac{(b-a)}{\mathcal{N}} f(x_j) - \frac{(b-a)}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} f(x_i) \right)^2, \tag{9}$$

in the *importance sampling* case we have that

$$Var[\frac{f(x)}{\Pi(x)}] = \frac{1}{(\mathcal{N}-1)} \sum_{j=1}^{\mathcal{N}} \left(\frac{1}{\mathcal{N}} \frac{f(x_j)}{\Pi(x_j)} - \frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} \frac{f(x_i)}{\Pi(x_i)} \right)^2, \tag{10}$$

and, the more the ratio $\frac{f(x)}{\Pi(x)}$ becomes constant, the lower the variance becomes.

Metropolis–Hastings algorithm

Monte Carlo integration

To sample a general probability density $\Pi(x)$ we use the **Metropolis–Hastings algorithm** that uses a single (or multiple) Markov chains.
Within the Metropolis–Hastings algorithm $\Pi(x)$ the explicit normalization of $\Pi(x)$ can be unknown.

1. Initialize a *Markov chain* in a configuration x_0. Better an initial point where $\Pi(x)$ is large.
2. Propose a trial move from δx to a new position x' with transition probability $\mathcal{T}(x' \leftarrow x)$, so that the probability that a *walker* initially in x afterwards transitions in an infinitesimal volume dx' of the configurational space is equal to $dx'\mathcal{T}(x' \leftarrow x)$;
3. Accept the move with probability

$$\mathcal{A}(x' \leftarrow x) = \min\left\{\frac{\Pi(x')\mathcal{T}(x \leftarrow x')}{\Pi(x)\mathcal{T}(x' \leftarrow x)}, 1\right\}; \tag{11}$$

4. If the trial move is accepted x' becomes the new position of the *walker* and a new transition is proposed from there, else a new transition is proposed from the previous position x. (Repeat from point 2 until $\mathcal{N}$ samplings are obtained).

Notice that a single point x might be repeated during the sampling.

Metropolis–Hastings algorithm in practice

Monte Carlo integration

In practice a metropolis algorithm can be defined as

1. Set x_0 in $[a, b]$.
2. Chose a move with transition probability $\mathcal{T}(x' \leftarrow x) = \frac{1}{2\Delta}$, that is $x' = x + \Delta\eta$ where Δ a chosen constant and η a random number in the intervale [-1:1].
3. Compute the acceptance probability

$$\mathcal{A}(x' \leftarrow x) = \min\left\{\frac{\Pi(x')}{\Pi(x)}, 1\right\}; \tag{12}$$

   where $\mathcal{T}(x' \leftarrow x) = \mathcal{T}(x \leftarrow x') = \frac{1}{2\Delta}$, and accept the move if $\mathcal{A}(x' \leftarrow x) \geq 1$ or, if $\mathcal{A}(x' \leftarrow x) \geq \eta$ where η is a random number uniformly extracted in the interval $[0 : 1]$
4. If the trial move is accepted x' becomes the new position of the *walker* and a new transition is proposed from there, else a new transition is proposed from the previous position x.
5. Compute the value of $f(x)/\Pi(x)$ on the new (or old) configuration and store it to compute the averages.
6. Continue from point 2.

For a finite interval $[a, b]$ is a move is proposed outside the boundaries, apply Periodic Boundary conditions (more efficient), or refuse the move.

Detailed Balance

Monte Carlo integration

The Metropolis–Hastings algorithm satisfies the *detailed balance principle* after an equilibration period, that is that every value of x generated is distributed according to $\Pi(x)$.
This equilibrium condition is expressed as:

$$dx' \mathcal{A}(x' \leftarrow x)\mathcal{T}(x' \leftarrow x)\Pi(x)dx = dx\mathcal{A}(x \leftarrow x')\mathcal{T}(x \leftarrow x')\Pi(x')dx'$$

so that:

$$\frac{\Pi(x')}{\Pi(x)} = \frac{\mathcal{A}(x' \leftarrow x)\mathcal{T}(x' \leftarrow x)}{\mathcal{A}(x \leftarrow x')\mathcal{T}(x \leftarrow x')}.$$

By substituting the acceptance probability defined in the previous slide $\mathcal{A}(x' \leftarrow x) = \frac{\Pi(x')\mathcal{T}(x \leftarrow x')}{\Pi(x)\mathcal{T}(x' \leftarrow x)}$ we have

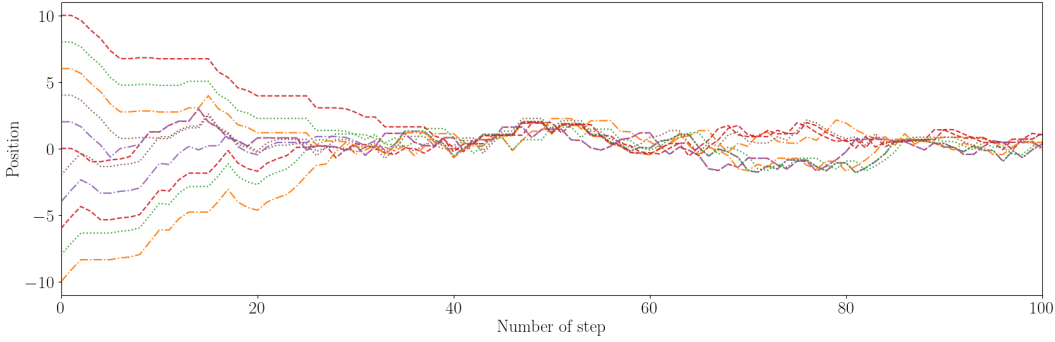
$$\frac{\Pi(x')}{\Pi(x)} = \frac{\Pi(x')}{\Pi(x)}$$

and this guarantees that the random walk is actually sampling the correct probability density $\Pi(x)$ and so that the *detailed balance principle* is satisfied.

Slide on Thermalization

Monte Carlo integration

Repeating dynamics with the same random seed, same step amplitude Δ and starting from different positions, sampling a Gaussian distribution with Standard Deviation $\sigma = 1$:

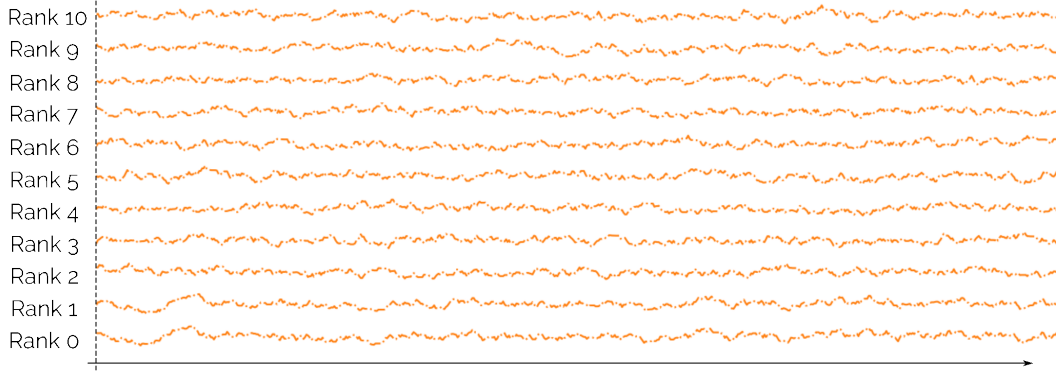


We see that the random walks are 'correlated withing each other' (because the random seeds are the same and thus the proposed moves are the same, only the starting points change), notice that each walker has a different thermalization time (because they require a different number o steps to start sampling the high probability region within σ, i.e. $x \in [-1 : 1]$).

Parallelizing with MPI: Embarrassingly Parallel

Monte Carlo integration

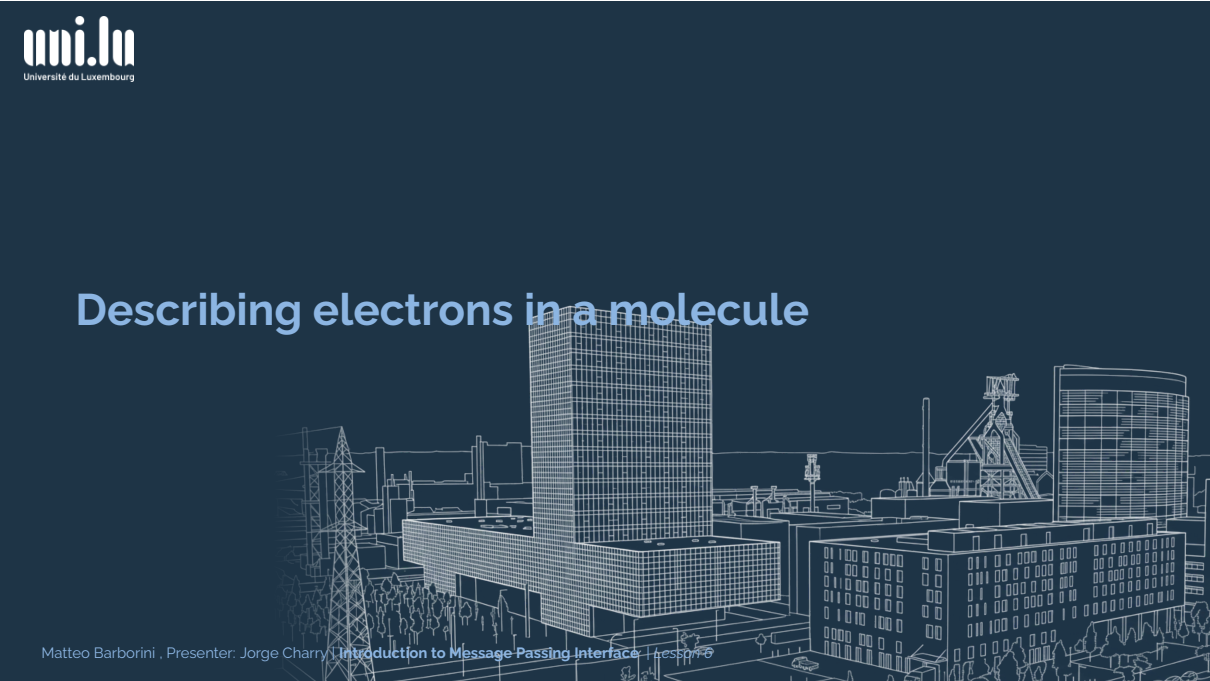
Using MPI, each task performs an independent random walk, using a different initial position and initial random seed.



each MPI task collects the necessary information.
The communication of the information is limited to the rank that needs to compute the final statistical averages.

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Describing electrons in a molecule



Time independent Schrödinger equation

Describing electrons in a molecule

The full electronic Schrödinger equation

$$\left[-\sum_{i=1}^{N} \frac{1}{2} \nabla_{\mathbf{r}_i}^2 - \sum_{a,i=1}^{M,N} \frac{Z_a}{|\mathbf{R}_a - \mathbf{r}_i|} + \sum_{i>j=1}^{N} \frac{1}{|\mathbf{r}_j - \mathbf{r}_i|} \right] \Psi_0(\bar{\mathbf{x}}) = E_0(\bar{\mathbf{R}}) \Psi_0(\bar{\mathbf{x}})$$

is not solvable analytically because of the coulomb interaction between electrons.

Time independent Schrödinger equation

Describing electrons in a molecule

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How can we approximately calculate the ground state properties?

Computational Methods	Scaling	Electronic Correlation	Solution based on
DFT	N^3	Depends on functional	Charge density
HF	N^3	Mean field	Wave function
CC, CI	$N^{5\sim7}$	Explicit correlation	Wave function
MP	$N^{5\sim7}$	Explicit correlation	Perturbation theory
QMC	$N^{3\sim4}$	Explicit correlation	Wave function

Time independent Schrödinger equation

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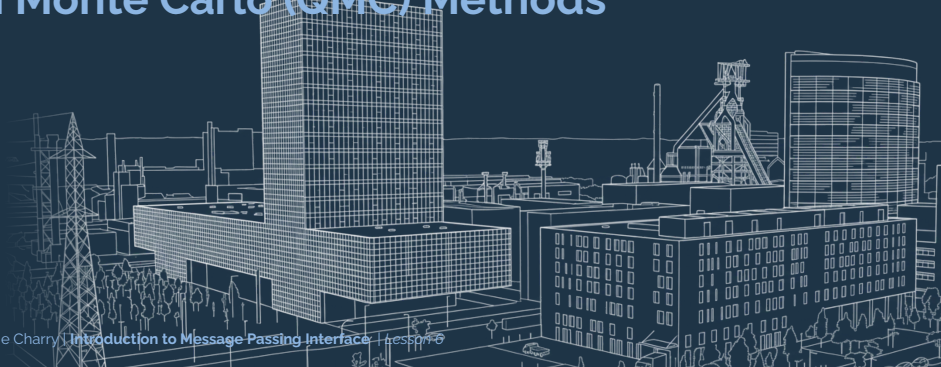
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QMC	$N^{3\sim4}$	Explicit correlation	Wave function

Quantum Monte Carlo methods ?

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Quantum Monte Carlo (QMC) Methods



Quantum Monte Carlo (QMC)[1]

⇓

Stochastic techniques

to

evaluate the mean values of physical observables

over

trial wave functions

Advantages:

- Good Scaling with the number of electrons $N^{3\sim4}$
- easy parallelization of the algorithms
- complex wave functions with explicit correlation

Disadvantages:

- Large computational prefactor due to the Stochastic nature of the method
- stochastic error

Variational Monte Carlo (VMC)

Quantum Monte Carlo (QMC) Methods

We consider a trial wave function $\Psi_T(\bar{\alpha}, \bar{\mathbf{r}})$ the depends on the set of parameters $\bar{\alpha}$.

The Variational Monte Carlo[1] method integrates stochastically the energy functional.

$$E\left[\Psi_T(\bar{\alpha}\})\right] = \frac{\int_V \Psi_T^*(\bar{\alpha}, \bar{\mathbf{r}}) \hat{H}_e \Psi_T(\bar{\alpha}, \bar{\mathbf{r}}) d\bar{\mathbf{r}}}{\int_V |\Psi_T(\bar{\alpha}, \bar{\mathbf{r}})|^2 d\bar{\mathbf{r}}}$$

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To do so the integrand of the energy functional is rewritten as:

$$E\left[\Psi_T\right] = \int_V \frac{|\Psi_T(\bar{\mathbf{r}})|^2}{\int_V |\Psi_T(\bar{\mathbf{r}})|^2 d\bar{r}} \frac{\hat{H}_e \Psi_T(\bar{\mathbf{r}})}{\Psi_T(\bar{\mathbf{r}})} d\bar{\mathbf{r}} = \int_V E_{loc}(\bar{\mathbf{r}}) p(\bar{\mathbf{r}}) d\bar{\mathbf{r}}$$

the product of two function:

Local Energy

$$E_{loc}(\bar{\mathbf{r}}) = \frac{\hat{H}_e \Psi_T(\bar{\mathbf{r}})}{\Psi_T(\bar{\mathbf{r}})}$$

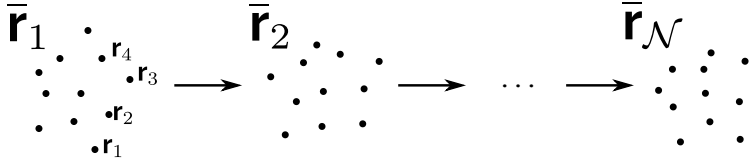
Sampling probability

$$p(\bar{\mathbf{r}}) = \frac{|\Psi_T(\bar{\mathbf{r}})|^2}{\int_V |\Psi_T(\bar{\mathbf{r}})|^2 d\bar{\mathbf{r}}}$$

Variational Monte Carlo (VMC)

Quantum Monte Carlo (QMC) Methods

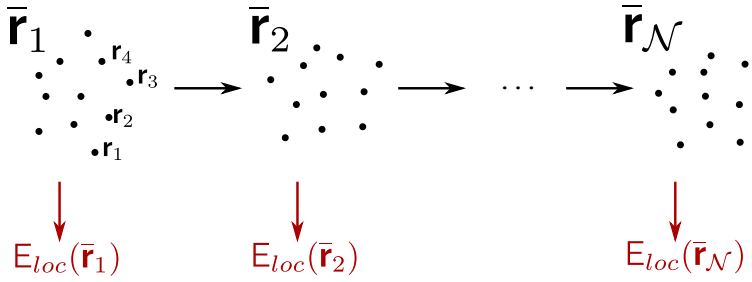
Random walk with $\mathcal{N}$ electronic configurations extracted with probability $p(\bar{\mathbf{r}}) \propto |\Psi_T(\bar{\mathbf{r}})|^2$



Variational Monte Carlo (VMC)

Quantum Monte Carlo (QMC) Methods

Random walk with $\mathcal{N}$ electronic configurations extracted with probability $p(\bar{\mathbf{r}}) \propto |\Psi_T(\bar{\mathbf{r}})|^2$

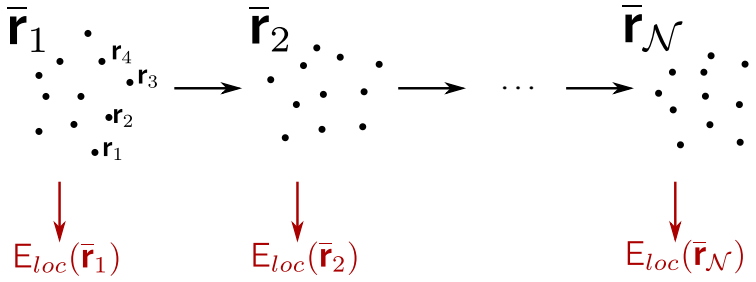


For each configuration we compute $E_{loc}(\bar{\mathbf{r}}) = \frac{\hat{H}_e \Psi_T(\bar{\mathbf{r}})}{\Psi_T(\bar{\mathbf{r}})}$

Variational Monte Carlo (VMC)

Quantum Monte Carlo (QMC) Methods

Random walk with $\mathcal{N}$ electronic configurations extracted with probability $p(\bar{\mathbf{r}}) \propto |\Psi_T(\bar{\mathbf{r}})|^2$



For each configuration we compute $E_{loc}(\bar{\mathbf{r}}) = \frac{\hat{H}_e \Psi_T(\bar{\mathbf{r}})}{\Psi_T(\bar{\mathbf{r}})}$

The estimation of the energy functional corresponds to the mean value of the Local energies:

$$E[\Psi_T] \approx \langle E_{loc} \rangle_{\mathcal{N}} \pm \sqrt{\frac{\langle E_{loc}^2 \rangle_{\mathcal{N}} - \langle E_{loc} \rangle_{\mathcal{N}}^2}{\mathcal{N}}}$$

Error decreases has $1/\sqrt{N}$ and the variance is zero for the exact wave function

Monte Carlo scalability in QMeCha

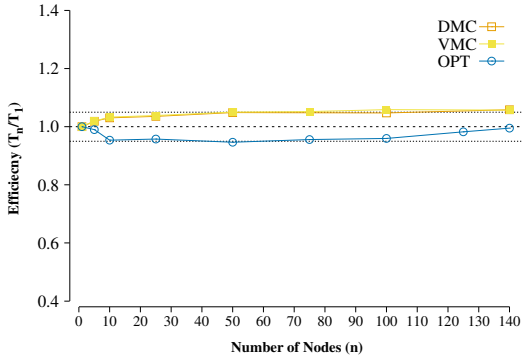
Quantum Monte Carlo (QMC) Methods

In QMeCha we showed a fixed molecular system, a cluster $(H_2O)_{30}$ of water molecules.

We use a number of independent Monte Carlo integration instances that are proportional to #Nodes × #MPI_tasks_per_node ×#OpenMP_threads number of processes.

Increasing the number of independent samplings corresponds to the increase of the problem.

We have nearly perfect weak scaling.



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Exercise: Monte Carlo Integration



Monte Carlo Integration of a one dimensional function

Exercise: Monte Carlo Integration

Consider the function $f(x) = x^2 e^{-x^2}$.

Exercise 1. Integrate the function in the interval $[-1:1]$ using the simplest Monte Carlo method (extracting $\mathcal{N}$ configurations with uniform probability in the interval [-1:1]). (Use the MT19937 pseudo random generator). Parallelize the integration with MPI_Send and MPI_Recv and MPI_Gather.

Q1 How does the error converge with respect to the number of samplings? How the variance and how the mean value?

Exercise 2. Choosing the distribution $\Pi(x) = \frac{1}{N} e^{-x^2}$ where N is the normalizing constant in $[-1:1]$ equal to $N = 1.493648265624854$, compute the same integral with importance sampling (Metropolis-Hastings algorithm). Remember to thermalize the random walk.

Q2 Is the variance of this estimation lower in this case then in Exercise 1? How fast does the estimation of the integral converge as a function of the number of samplings?

Q3 Using the MPI_Wtime function estimate the Efficiency of the algorithm as a function of MPI tasks (Increase the number of sampling proportional to the number of tasks)? Keeping the number of samplings (dimension of the problem) fixed, what is the Speed up increasing the number of MPI tasks?